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Write a Syntax for loop inside while loop

Syntax:-

initialization of while loop

while (Condition): # outer loop

for var in range(): # inner loop

statements of inner loop

statements of outer loop

incrementation/decrementation of while loop

Using this above Syntax write a program to print like #

```
* * * * *
*      *
*      *
*      *
* * * * *
```

i = 1

while (i <= 5):

for j in range(1, 6, 1):

if (i == 1 or i == 5 or j == 1
or j == 5):

Print ("*", end = " ")

else:

Print (" ", end = " ")

Print()

i = i + 1

Logic:-

• Outer loop (while) → controls the rows (i = 1 to 5)

• Inner loop (for) → controls the columns (j = 1 to 5)

At each position (i, j) in the grid:

1) Check if the current position lies on the border of the square:

- i == 1 → top row
- i == 5 → bottom row
- j == 1 → leftmost column
- j == 5 → rightmost column

1) If on the border $\rightarrow$ print *
 2) otherwise (inside the square) $\rightarrow$ print a blank space
 After finishing all column (j), move to the next line (row)
 Imagine the grid as coordinates (row, column):

• Row = i, column = j

• Borders $\rightarrow$ when row = 1, row = 5, col = 1 or col = 5

• otherwise, print a ~~star~~ space

So the code places * around the edges, spaces inside

(1,1) (1,2) (1,3) (1,4) (1,5) $\rightarrow$ row 1 $\rightarrow$ all stars
 (2,5) $\rightarrow$ row 2 $\rightarrow$ stars only at col 5

(2,1)

(3,5) $\rightarrow$ row 3 $\rightarrow$ same

(3,1)

(4,5) $\rightarrow$ row 4 $\rightarrow$ same

(4,1)

(5,1) (5,2) (5,3) (5,4) (5,5) $\rightarrow$ row 5 $\rightarrow$ all stars

Write a Syntax of while loop inside for loop

for var in range(): # outer loop

initialization of while loop

while (condition): # inner loop

Statements of inner loop

incrementation / decrementation of while loop

Statements of outer loop

Write a Program to print like this

```

  1
 2 2
3 3 3
4 4 4 4
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
  
```

for i in range(1, 6, 1):

j = 5

while (i < j):

Print(" ", end=" ")

j = j - 1

k = 1

while (k <= i):

Print(i, end=" ")

k = k + 1

Print()

for i in range(4, 0, -1):

j = 5

while (j > i):

Print(" ", end=" ")

j = j - 1

k = 1

while (k <= i):

Print(i, end=" ")

k = k + 1

logic of 1st loop: Print()

Row control: i goes from 1 to 5 → 5 rows printed (top half of diamond)

Spaces (while (i < j)):

- Starts with more spaces, decreases each row
- Row 1 → 4 spaces, Row 2 → 3 spaces ... Row 5 → 0 spaces
- This shifts the numbers to the right, forming the Pyramid shape.

Numbers (while (k <= i)):

- Prints i, exactly i times
- Row 1 → 1
- Row 2 → 22
- Row 3 → 333
- Row 4 → 4444
- Row 5 → 55555

So this block makes the upper pyramid

logic of 2nd loop

Row control: i goes from 4 $\rightarrow$ 1 (descending)

Spaces (while ($j > i$)):

- Row 4 $\rightarrow$ 1 Space
- Row 3 $\rightarrow$ 2 Spaces
- Row 2 $\rightarrow$ 3 Spaces
- Row 1 $\rightarrow$ 4 Spaces

Numbers (while ($k \leq i$)):

- Prints i , exactly: times.
- Row 4 $\rightarrow$ 4 4 4 4
- Row 3 $\rightarrow$ 3 3 3
- Row 2 $\rightarrow$ 2 2
- Row 1 $\rightarrow$ 1

- First loop $\rightarrow$ builds the top pyramid (increasing rows, 1 to 5)
- Second loop $\rightarrow$ builds the bottom pyramid (decreasing rows, 4 to 1)
- Together $\rightarrow$ forms a diamond of numbers, symmetric around the center row.

The Program's logic is to first print spaces to align numbers centrally, then print the row number (i) multiple times to form pyramids, and finally combine an upright pyramid with an inverted pyramid to form a diamond.

$\rightarrow$ Step by Step for the first 2 rows so you can see how variables i , j , and k behave

first for loop $\rightarrow$ Top Half

Row ($i=1$)

$j=5$

while ($i \leq j$): # while ($i \leq 5$)

- iteration 1: $i=1, j=5 \rightarrow$ condition true $\rightarrow$ print " " $\rightarrow j=4$
 - iteration 2: $i=1, j=4 \rightarrow$ true $\rightarrow$ print " " $\rightarrow j=3$
 - iteration 3: $i=1, j=3 \rightarrow$ true $\rightarrow$ print " " $\rightarrow j=2$
 - iteration 4: $i=1, j=2 \rightarrow$ true $\rightarrow$ print " " $\rightarrow j=1$
 - iteration 5: $i=1, j=1 \rightarrow$ false $\rightarrow$ stop
- On 4 spaces are printed

Now the numbers:

$k=1$
 $\text{while}(k \leq i): \# \text{while}(1 \leq 1)$
• iteration 1: $k=1 \rightarrow \text{print } 1 \rightarrow k=2 \rightarrow \text{condition false}$
So one 1 is printed

Row: o/p
1

Row 2 ($i=2$)

$j=5$
 $\text{while}(i \leq j): \# \text{while}(2 \leq 5)$
• iteration 1: $i=2, j=5 \rightarrow \text{true} \rightarrow \text{print " " } \rightarrow j=4$
• iteration 2: $i=2, j=4 \rightarrow \text{true} \rightarrow \text{print " " } \rightarrow j=3$
• iteration 3: $i=2, j=3 \rightarrow \text{true} \rightarrow \text{print " " } \rightarrow j=2$
• iteration 4: $i=2, j=2 \rightarrow \text{false} \rightarrow \text{stop}$
So 3 spaces are printed

Now the numbers:

$k=1$
 $\text{while}(k \leq i): \# \text{while}(1 \leq 2)$
• iteration 1: $k=1 \rightarrow \text{print } 2 \rightarrow k=2$
• iteration 2: $k=2 \rightarrow \text{print } 2 \rightarrow k=3 \rightarrow \text{condition false}$
So 2 2 is printed

Row 2 o/p

2 2

Go far (after 2 rows)

2 2

Note:

* which are the variables we called inside. Funct are called local variables.

* which are the variables we called outside. Funct are called Global variables.
* which are the variables we called inside the function that can be printed inside that function only if you try to print that variables outside the function you will get Error